Test Plan for the Smart Greenhouse Monitoring System

All tests are run manually from the terminal. Compile first with javac -d bin Java/*.java, then run with java -cp bin Main. The CSV file is data.csv in the working directory unless a row says otherwise. The baseline data.csv used for these tests contains the header plus the 5 rows shipped with the project (TMP332, HMD221, SOIL88, LGT901, JAS116). Safe ranges checked by the program are temperature 18-30, humidity 40-70, soilMoisture 30-60, light 300-1200. Of the baseline data, only JAS116 (temperature 55.5) is outside its safe range.

Test ID	Test Description	Input	Expected Output	Actual Output	Pass/Fail
CSV LOADING					
T01	Load a valid CSV file	Baseline data.csv (header + 5 well-formed rows)	SensorArray is populated with all 5 readings. No Warning: lines printed. Program proceeds straight to the main menu.	No warnings printed. Main menu shown immediately. Subsequent stat query [1] -> [1] reports Total: 5.	Pass
T02	Load a missing CSV file	data.csv removed from the working directory	Error Error loading file 'data.csv': ... printed. Empty SensorArray returned. Program continues to the menu.	Error loading file 'data.csv': data.csv (No such file or directory) printed, main menu shown.	Pass
T03	Load a CSV with malformed rows	data.csv rewritten with: (a) a row missing the value field, (b) a row with day = abc, (c) a row with sensorType = wind	A Warning: skipped invalid row N: ... line is printed for each bad row. Remaining valid rows still load.	Three warnings printed for rows 3, 4, and 6. Subsequent Total: 2 confirms the two well-formed rows still loaded.	Pass
MENU INPUT VALIDATION					
T04	Non-numeric input at main menu	At choice: type abc and press Enter	Error Enter a number between 1 and 6. printed. Prompt repeats.	Enter a number between 1 and 6. printed, prompt repeats.	Pass
T05	Out-of-range integer at main menu	At choice: type 9	Enter a number between 1 and 6. printed. Prompt repeats.	Enter a number between 1 and 6. printed, prompt repeats.	Pass
T06	Empty input at main menu	At choice: press Enter with nothing typed	Enter a number between 1 and 6. printed. Prompt repeats.	Enter a number between 1 and 6. printed, prompt repeats.	Pass
T07	Negative integer at main menu	At choice: type -1	Enter a number between 1 and 6. printed. Prompt repeats.	Enter a number between 1 and 6. printed, prompt repeats.	Pass
STATISTICS CALCULATIONS					
T08	Total readings (entire greenhouse)	Main menu [1], then sub-menu [1]	Total: 5 (matches CSV row count).	Total: 5	Pass
T09	Average value	Main menu [1], then sub-menu [2]	Average: 213.21999999999997 ((27.4 + 61.2 + 42.0 + 880.0 + 55.5) / 5 = 213.22).	Average: 213.21999999999997	Pass
T10	Minimum value	Main menu [1], then sub-menu [3]	Minimum: 27.4 (smallest value across all readings).	Minimum: 27.4	Pass
T11	Maximum value	Main menu [1], then sub-menu [4]	Maximum: 880.0 (largest value across all readings).	Maximum: 880.0	Pass
T12	Out-of-range count	Main menu [1], then sub-menu [5]	Number outside safe range: 1 (only JAS116 at 55.5 is outside the temperature 18-30 band).	Number outside safe range: 1	Pass
T13	Out-of-range percentage	Main menu [1], then sub-menu [6]	Percent outside safe range: 20.0 ((1 / 5) * 100).	Percent outside safe range: 20.0	Pass
FILTERING					
T14	Filter by zone	Main menu [2], zone [1] (ZoneA), then sub-menu [7]	Stats sub-menu operates only on ZoneA rows (TMP332 and SOIL88): Total: 2, Average: 34.7, Minimum: 27.4, Maximum: 42.0, Number outside safe range: 0, Percent outside safe range: 0.0.	All six lines printed exactly as expected.	Pass
T15	Filter by sensor type	Main menu [3], type [1] (temperature), then sub-menu [7]	Stats sub-menu operates only on temperature rows (TMP332 27.4 and JAS116 55.5): Total: 2, Average: 41.45, Minimum: 27.4, Maximum: 55.5, Number outside safe range: 1, Percent outside safe range: 50.0.	All six lines printed exactly as expected.	Pass
ADD READING					

Test ID	Test Description	Input	Expected Output	Actual Output	Pass/Fail
T16	Add a valid reading	Main menu [4], sensor ID NEW001, type [1] (temperature), zone [1] (ZoneA), value 22.5, day 15, month 5, year 2026, hour 14, minute 30	Reading appended. Console prints Added: [15/05/2026 14:30] NEW001 (temperature, ZoneA): 22.5. data.csv is rewritten with the new row appended at the bottom.	Added: [15/05/2026 14:30] NEW001 (temperature, ZoneA): 22.5 printed. Inspecting data.csv after exit confirms the new row appended at the end.	Pass
T17	Add reading with invalid sensor type selection	After sensor ID, at type: type 9 (out of the 1-4 range), then type 1	Enter a number between 1 and 4. printed. Prompt repeats until a valid selection. (Note: the type selector is a numbered list, so free-text invalid types are not reachable - invalid input here means a number outside the printed range.)	Enter a number between 1 and 4. printed, then prompt re-displayed and 1 accepted. Reading added successfully on retry.	Pass
T18	Add reading with invalid date field	At day (1-31): type 32, then 15	Enter a number between 1 and 31. printed. Prompt repeats until a valid day.	Enter a number between 1 and 31. printed, prompt repeated, 15 accepted.	Pass
DELETE READING					
T19	Delete a valid reading	Main menu [5], then index 2 from the printed list	The reading at index 2 (HMD221) is removed. Console prints Deleted: [12/03/2024 14:30] HMD221 (humidity, ZoneB): 61.2. data.csv is rewritten without that row.	Deletion message printed verbatim. data.csv inspected after exit contains only the four remaining readings.	Pass
T20	Delete from an empty array	Load an empty CSV (header only), then main menu [5]	No readings to delete. printed. User returned to the main menu without an exception.	No readings to delete. printed, main menu re-shown.	Pass
LOG FILE					
T21	Log file session markers	Run the program against the baseline CSV, navigate menus, exit via [6]	logged_data contains starting session... at the top of the run's block and ending session at the bottom.	Confirmed by inspecting logged_data - first line of the run is starting session..., last line before the trailing blank line is ending session.	Pass
T22	Log file captures Logger output	Run the program, navigate through several menu options	Every line emitted via Logger.log (menu screens, stat results, error messages, Added: / Deleted: / No readings to delete. lines) appears in logged_data in the order it was printed. Inline input prompts (choice: , stat: , day (1-31): , etc.) are written with System.out.print directly and therefore do not appear in the log; this is the intended design.	Side-by-side comparison of console output and logged_data for a sample run confirmed every Logger.log line is present in the log in order. Prompts are absent from the log, as expected.	Pass